

BACKGROUND

Ellexacaftor-tezacaftor-ivacaftor (ELX/TEZ/IVA) is a modulator that treats the underlying cause of cystic fibrosis (CF) in patients with at least one F508del mutation in the CFTR gene. According to the CF Foundation Patient Registry, 84.7% of genotyped CF patients have at least one copy of the F508del mutation (CFFPR Annu Data Rep. 2018). With this breakthrough therapy, there is a need to evaluate the effect of a pharmacy clinical services model on economic outcomes.

OBJECTIVE

To determine overall cost avoidance associated with a multidisciplinary pharmacy-based clinical services program for CF patients initiating ELX/TEZ/IVA therapy.

METHODS

The primary objective was to determine cost avoidance associated with interventions performed for CF patients initiating ELX/TEZ/IVA. The secondary objective was to evaluate participation rates for counseling. Patients included in the data were those who received an initial dispense of ELX/TEZ/IVA between 10/21/19 and 4/21/20. Patients received a pharmacist-led chart review followed by a counseling call. Pharmacists also initiated medication reconciliation requests when needed. About one week after shipment, patients received a nurse call to assess presence of adverse drug events (ADE) with ELX/TEZ/IVA. Data included interventions for therapy consultation, ADE and drug-drug interaction (DDI) detection, and medication reconciliation. Average cost avoidance was used to determine potential cost savings associated per intervention type. According to previous research, cost avoidance per intervention was estimated at \$47.89, \$276.12, \$276.12, and \$30.12, respectively (Schmidt L, et al. Am J Health-Syst Pharm 2017; 74: e76-e82).

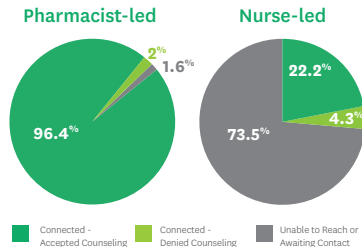
\$91.6K Total Cost Avoidance

Clinical Interventions		
Intervention Type	Intervention Conducted	Total Cost Avoidance
Pharmacist-led		
Therapy Consultation	910	\$43,579.90
Drug-Drug Interaction Detection	5	\$1,380.60
Medication Reconciliation	331	\$9,969.72
Nurse-led		
Adverse Drug Event Detection	133	\$36,723.96

RESULTS

A total of 944 patients met inclusion criteria. There were 910 pharmacist-led therapy consultations, 331 medication reconciliation requests, and 5 DDIs identified, with an associated cost avoidance of \$43,579.90, \$9,969.72, and \$1,380.60, respectively. There were 209 nurse-led follow-up calls, 133 of which identified an ADE, associated with \$36,723.96 cost avoidance. Pharmacist-led and nurse-led interventions had a connection rate of 96.40% and 22.19%, with 2% and 4.3% of patients declining counseling, respectively.

Connection Rates for Pharmacist-led and Nurse-led Intervention



CONCLUSIONS

Integration of multidisciplinary clinical programs is an effective way to enhance patient care when initiating new therapy. The low percentage of patients who declined counseling suggests patients' perceived value of clinician counseling in pharmacy settings.